Climate-Aware Capital Allocation Under NGFS Scenarios

Final Report – Undergraduate Thesis

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Glossary

NGFS – Network for Greening the Financial System.
DAPS – Delayed Action and Physical Stress.
DiRe – Diverging Realities.
HWTP – Higher-Weight Transition Pathway.
SWUC – Sudden Wake-Up Call.
PD – Probability of Default.
VaR – Value at Risk.
ES – Expected Shortfall.
EPL – Expected Physical Loss.
LGD – Loss Given Default.
PCA – Principal Component Analysis.
SVD – Singular Value Decomposition.

Structured Abstract

Background: Climate stress can change not only expected credit losses but also which firms remain attractive to fund, how concentrated the funded portfolio becomes, and how severe tail losses remain after allocation adapts. The source paper by Bourgey, Gobet, and Jiao studies a fixed-exposure climate-credit portfolio, and Marina Palaisti extends that direction into a capital-allocation setting.
Objective: This thesis asks how climate scenarios change firm-level default risk, capital allocation, concentration, and portfolio tail losses under a synthetic but structured bank-style portfolio.
Method: The thesis uses NGFS short-term scenario workbooks, builds region-sector stress objects, assigns a synthetic portfolio to that grid, estimates default via structural first-passage Monte Carlo, allocates capital with a capped proportional rule, and evaluates portfolio outcomes with loss distributions, VaR, Expected Shortfall, and sensitivity analysis. A three-layer comparison is used: source-paper reproduction, paper-aligned fixed-exposure bridge, and full endogenous allocation.
Results: The most favorable scenario, HWTP, produces the safest funded portfolio on average and in the tail, but it also creates the most concentrated allocation. SWUC produces the worst tail losses by a clear margin. Endogenous allocation reduces some average physical-risk uplift relative to fixed exposures, but it does not remove severe tail exposure. The region-aware extension also improves the distinction between DAPS and DiRe.
Conclusion: Climate-aware capital allocation changes the portfolio decision itself, not only the measured risk of a fixed book. The thesis shows that scenario design, geographic structure, and endogenous allocation all matter for how banks should interpret climate stress.

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1 Introduction

Climate risk matters to banks because it affects both firm fundamentals and portfolio decisions. A climate stress model is therefore more useful when it does more than report stressed default probabilities. It should also show how scenario design changes funding choices, portfolio concentration, and the upper tail of losses. This thesis addresses that problem using a structural credit-and-allocation framework driven by NGFS scenario inputs.

The focus of the thesis is a synthetic but structured portfolio setting. The firms are simulated, but the scenario layer is drawn from real NGFS short-term climate scenario workbooks. This separation is deliberate. The project is not trying to forecast the exact losses of an actual bank book. Instead, it studies how real climate stress data propagates through a transparent allocation and risk framework.

The main research question is: *How do climate scenarios change firm-level default risk, portfolio allocation, concentration, and tail losses?* The thesis answers this question by starting from the structural climate-credit methodology in Bourgey, Gobet, and Jiao (2024), retaining the allocation perspective introduced by Palaisti (2025), and then extending the workflow with NGFS-driven scenario mapping, a region-aware synthetic portfolio, structural first-passage Monte Carlo default estimation, and portfolio loss analysis.

The key results are straightforward. HWTP produces the safest funded portfolio on average and in the tail. SWUC is the harshest scenario in the upper tail. DAPS and DiRe remain closer than ideal, but the region-aware extension improves their separation. The novelty of the work lies in combining a structural Monte Carlo engine, NGFS scenario data, region-sector stress transmission, and endogenous capital allocation in one reproducible workflow. The significance is that the thesis moves the framework from a fixed-exposure stress test toward a bank-style portfolio decision problem.

The rest of the report is structured as follows. Section 2 reviews the source framework and related work. Section 3 states the overall research goal and objectives. Section 4 explains the methodology. Section 5 presents the results. Section 6 discusses threats to validity, implications, limitations, and generalisability. Sections 7 and 8 conclude with findings, future work, and lessons learnt.

2 Background and Related Work

2.1 Structural Climate-Credit Modeling

The source paper by Bourgey, Gobet, and Jiao (2024) provides the structural starting point for this thesis. In that framework, climate stress affects firm dynamics and the default condition, portfolio loss can be written as exposure times a default indicator, and factor compression makes portfolio loss dependence tractable. The importance of that paper for this thesis is not just the equations themselves, but the modeling philosophy: climate risk enters firm behavior and then propagates into portfolio loss.

2.2 Climate-Aware Capital Allocation

Palaisti (2025) extends the structural direction into a capital-allocation setting. This is important because a bank does not only hold a stressed portfolio passively. It also changes its funding decisions. That shift from fixed exposure to endogenous allocation is central to the present thesis. The allocation question introduces a practical decision layer: which firms remain attractive to fund once climate stress changes default risk?

2.3 NGFS Scenario Framework

The scenario layer in this thesis comes from the NGFS short-term climate scenario release and the related NGFS guidance for climate scenario analysis (Network for Greening the Financial System [NGFS], 2025a, 2025b, 2025c). The direct-impact workbooks drive the physical-style scenarios DAPS and DiRe, while the CLIMACRED workbook drives the transition and repricing scenarios HWTP and SWUC. These scenario families matter because they provide different stress mechanisms rather than just different parameter values.

2.4 Analysis and Research Gap

Across these sources, a clear research gap appears. The source paper gives a strong structural benchmark but stays in a fixed-exposure setting. Palaisti introduces allocation, but the earlier implementation remained more stylized in its scenario layer and did not yet fully exploit NGFS region-sector information. The key gap addressed here is therefore a combined one: how to connect a structural credit model, real NGFS scenario data, a region-aware synthetic portfolio, and endogenous capital allocation in one coherent workflow.

This gap relates directly to the core work of the thesis. The thesis does not simply reproduce earlier results. It builds a comparison ladder that separates three effects: the source-paper benchmark, the updated risk engine under fixed exposures, and the final effect of turning allocation on. That makes it possible to attribute differences more carefully.

3 Research Goal and Objectives

The overall goal of the thesis is to evaluate how climate scenarios change both credit risk and capital allocation decisions in a structured portfolio setting.

The objectives are:

- **O1:** Build a consistent scenario layer from NGFS workbooks that can transmit stress into firm-level credit dynamics.
- **O2:** Estimate firm default probability using a structural first-passage Monte Carlo model rather than a post-hoc reduced-form shortcut.

- **O3:** Allocate capital endogenously across firms under climate stress and measure how allocation changes concentration and sector-region tilts.
- **O4:** Compare source-paper, fixed-exposure, and fully allocated results in order to isolate which differences come from the risk engine and which come from the allocation rule.

Together, O1–O4 support the overall goal by connecting real scenario inputs, structural credit modeling, capital allocation, and portfolio loss evaluation in a single thesis workflow.

4 Methodology

The thesis begins by reading three NGFS workbook sources: direct impacts for DAPS, direct impacts for DiRe, and CLIMACRED for HWTP and SWUC (NGFS, 2025a). The raw workbook rows are not used directly in the credit model. Instead, they are aggregated into region-sector stress rows. For DAPS and DiRe, the code maps productivity loss, production loss, and capital destruction into the main stress channels $(\theta_1, \theta_2, \theta_3)$, while also building a severity scaler, tail-intensity term, EPL fraction, and LGD stress term. For HWTP and SWUC, the same common stress interface is built from spread, bond-price, equity, and PD-adjustment fields.

The synthetic portfolio is then placed on a shared region-sector grid. This design choice matters for O1 and O3 because one of the weaknesses of the earlier sector-only version was that DAPS and DiRe looked too similar. Since DiRe is fundamentally about uneven regional outcomes, the thesis constructs firms on region-sector entities rather than on sectors alone.

Climate stress enters firm dynamics through transformed stress terms

$$\gamma_{i,s,j} = \frac{\theta_{scale,i,s}\theta_{i,s,j}}{2\beta_j(r + b_i - \alpha_j)},$$

and the stressed drift becomes

$$\mu^{\theta}_{i,s} = a_i - b_i p_{0,i} + \sum_{j=1}^{3} \omega_j \gamma_{i,s,j}.$$

The default barrier is anchored by

$$p^{ref} = 1 - e^{-\lambda_{ref} T},$$

with $\lambda_{ref} = 3\%$ and $T = 5$ years in the baseline setting.

For O2, default probability is estimated through structural first-passage Monte Carlo. The latent firm state is simulated over multiple time steps using common factors, idiosyncratic shocks, volatility scaling, and regime penalties. A default occurs if the simulated path crosses the stressed barrier at any time before maturity. The estimator is

$$\widehat{PD}_{i,s} = \frac{1}{M} \sum_{m=1}^{M} 1\{\tau^{(m)}_{i,s} \leq T\}.$$

For O3, capital allocation is computed from the score

$$score_{i,s} = \eta_i(1 - \widehat{PD}_{i,s}),$$

where η_i is a firm-level attractiveness term. The score is converted into portfolio weights through a capped proportional allocation rule with a diversification cap of 2% per firm.

For O4, the thesis evaluates results at three levels: source-paper benchmark reproduction, paper-aligned fixed-exposure bridge, and full allocation. Portfolio loss is then simulated using the allocated exposures, and the analysis reports mean loss, VaR, Expected Shortfall, sector and region allocation patterns, and sensitivity to the risk-free rate.

5 Results

5.1 Source-Paper Benchmark and Comparison Ladder

The source-paper benchmark is reproduced in paper-style form in order to create a stable reference point. It shows the original fixed-exposure logic clearly: when physical risk is added, the portfolio loss distribution shifts materially to the right, especially in the hotter baseline scenarios.

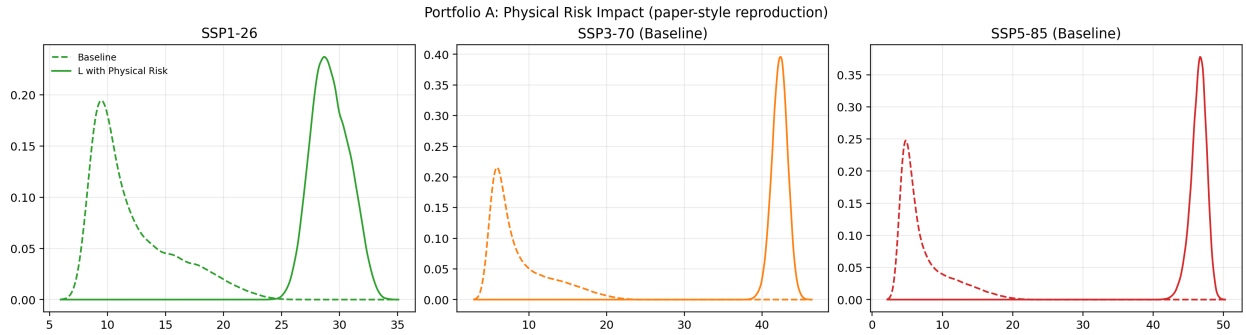

Figure 1: Paper-style reproduction of the source benchmark. Physical risk shifts the fixed-exposure loss distribution to the right in all three SSP scenarios.

That benchmark is not the final thesis result. It is the first layer in the comparison ladder. The second layer keeps the paper-style exposure logic but replaces the stylized source scenario system with the thesis risk engine. The third layer turns allocation on and measures how endogenous funding changes the outcome.

5.2 Main Thesis Baseline

The final thesis baseline uses 5,000 synthetic obligors, the 2027 NGFS snapshot, a 5-year horizon, a 2% risk-free rate, a 3% reference hazard anchor, structural first-passage Monte Carlo PD estimation, and capped proportional allocation.

Table 1: Scenario Summary for the Final Thesis Baseline

Scenario	Mean PD	Capital-weighted PD	Effective obligors	Top-20 share	Top sector	Top region
DAPS	0.624	0.342	807.0	10.16%	Advanced Electric Appliances	Japan
DiRe	0.595	0.333	945.2	8.79%	CSS Bio	Greece
HWTP	0.549	0.212	278.6	22.11%	Biofuels	Japan
SWUC	0.669	0.375	803.9	10.22%	Advanced Electric Appliances	Greece

Table 1 addresses O3 directly. HWTP has the lowest capital-weighted PD, but it also produces the most concentrated funded portfolio. SWUC is clearly the harshest stress case. DAPS and DiRe remain closer than ideal, but the region-aware design prevents them from collapsing into the same output.

Table 2: Loss Metrics for the Final Thesis Baseline

Scenario	Mean loss	VaR95	VaR99	ES95	ES99
DAPS	10.83%	34.57%	44.06%	40.74%	49.45%
DiRe	10.58%	31.88%	40.52%	37.31%	45.21%
HWTP	6.76%	21.00%	29.26%	26.15%	33.58%
SWUC	13.27%	43.83%	58.12%	53.87%	71.53%

Table 2 addresses O4. HWTP is the safest scenario in both average loss and the tail. SWUC is the dominant crisis scenario. The gap between average and tail outcomes is one of the main findings of the thesis: ranking scenarios by mean PD alone would miss the severity of the upper tail.

Figure 2 reveals an important pattern. The scenarios are less separated around the middle of the loss distribution than in the upper tail. This is not a charting mistake. The minimum simulated loss is above zero in all scenarios, so the left-side mass is concentrated near small positive losses rather than at zero. The stronger separation emerges when systemic stress becomes severe, which is why SWUC looks most different in the tail.

5.3 Region-Aware Allocation and Sensitivity

The region-aware extension improves the meaning of DiRe because firms are no longer modeled as sector-only objects. The outputs are still synthetic rather than empirical, but the regional layer is now genuinely tied to scenario structure rather than being incidental.

The thesis also studies sensitivity to the risk-free rate. As r increases, capital-weighted PD generally declines, but concentration can rise, especially in HWTP. This matters for O3 because a safer-looking portfolio is not automatically a more diversified one.

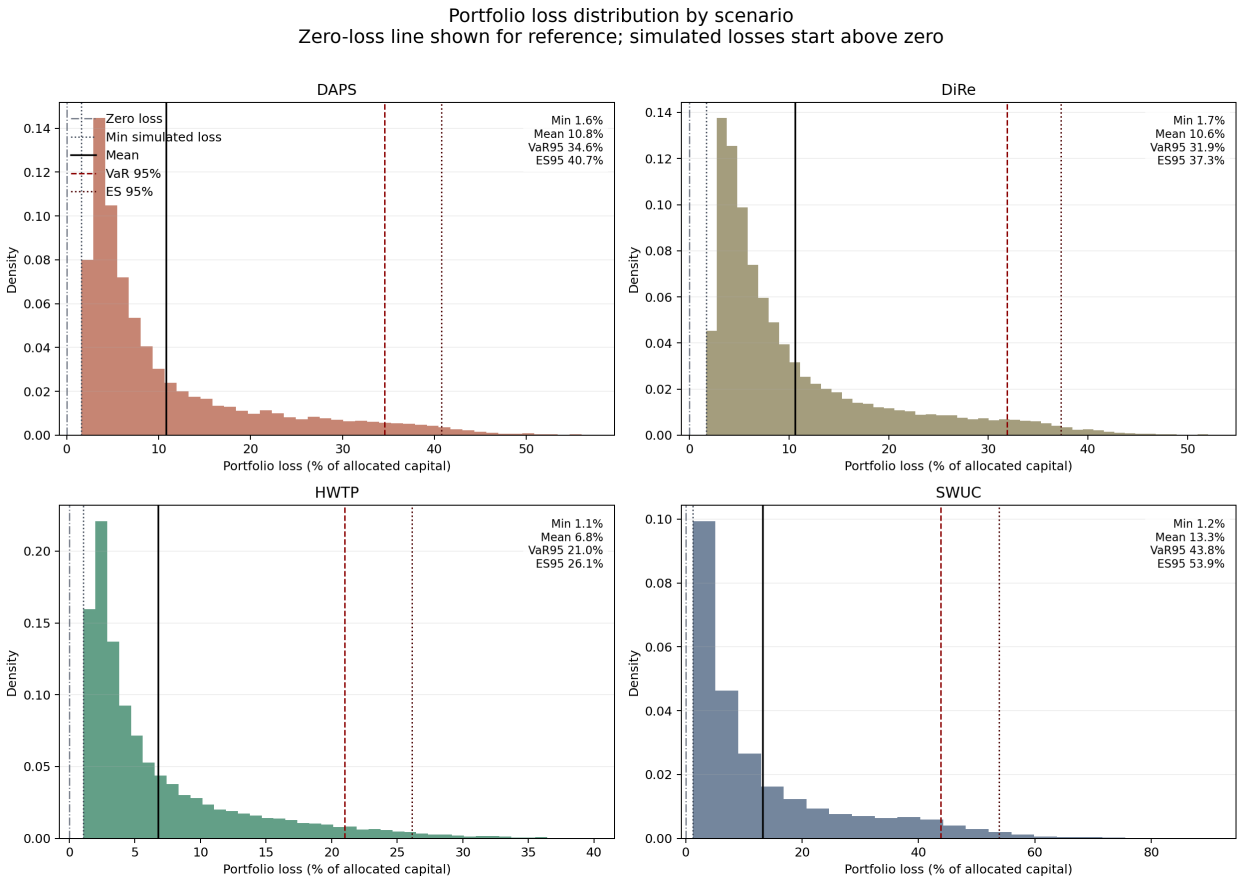

Figure 2: Final thesis loss distributions by scenario. The distributions share some central mass, but the tail behavior separates sharply, especially for SWUC.

6 Discussion

6.1 Threats to Validity

Several threats could affect the quality of the results. First, the firms are synthetic rather than drawn from a real bank book. Second, the stress descriptors are calibrated heuristically from NGFS rows rather than estimated from proprietary exposure histories. Third, the portfolio-loss engine remains stylized relative to a full network contagion model. These threats were partly mitigated by using a comparison ladder, by checking the source-paper benchmark separately, and by making the scenario-to-firm mapping explicit.

6.2 Implications of the Results

The thesis has implications for both research and practice. For research, it shows that a fixed-exposure climate-credit result can change materially once endogenous allocation is introduced. For practice, it suggests that climate stress testing should inform funding and concentration decisions,

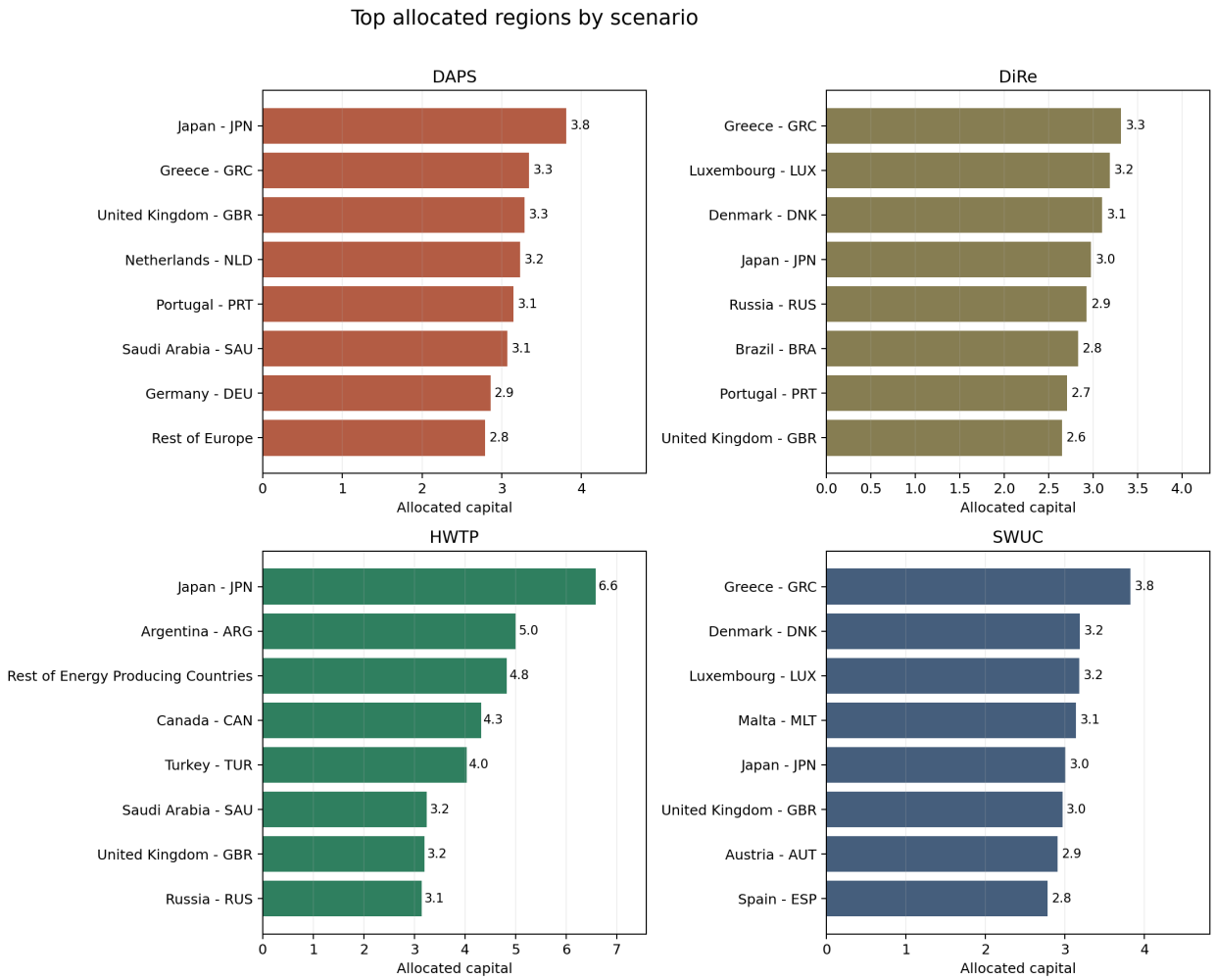

Figure 3: Top allocated regions under the final thesis baseline. Geography now affects the allocation outcome directly.

not only expected-loss accounting. In particular, the difference between HWTP and SWUC shows why banks should evaluate tail behavior separately from average risk.

6.3 Limitations of the Results

The main limitations are that the portfolio is synthetic, DAPS and DiRe are still not separated as strongly as desired, and the regional layer is more meaningful directionally than empirically. These limits do not invalidate the thesis. They define the scope of the work as a methodological and computational extension rather than as a production-ready bank implementation.

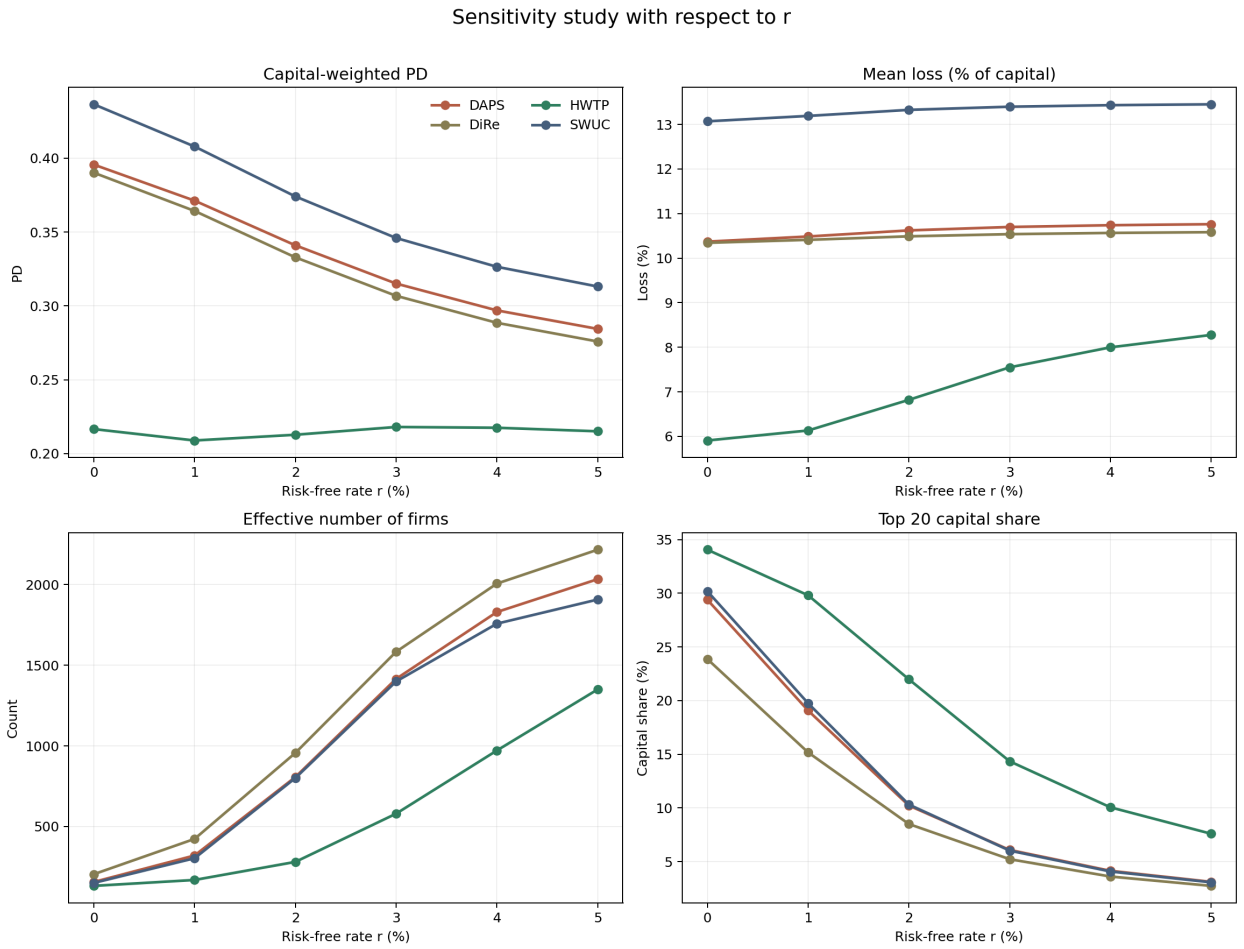

Figure 4: Sensitivity of the final thesis baseline to the risk-free rate. Average quality improves, but concentration can still rise.

6.4 Generalisability

The results are generalisable at the level of framework design rather than at the level of exact numeric outputs. The thesis shows how a structural Monte Carlo engine, NGFS scenario inputs, and endogenous allocation can be connected. However, the specific numbers should not be transferred directly to other portfolios without recalibrating the synthetic firms and the exposure structure.

7 Conclusions

This thesis studied how climate scenarios change default risk, capital allocation, concentration, and tail losses in a structured portfolio setting. The work addressed O1 by building a common NGFS stress interface, O2 by restoring structural first-passage Monte Carlo default estimation, O3 by allocating capital endogenously under diversification constraints, and O4 by separating benchmark, fixed-exposure, and allocated results through a comparison ladder.

The main conclusions are supported directly by the results in Section 5. Table 1 shows that HWTP produces the safest funded portfolio on average but also the most concentrated one. Table 2 and Figure 2 show that SWUC remains the harshest scenario in the upper tail. Figure 3 shows that the region-aware extension improves the scenario story by making geography matter directly. Together, these results show that climate-aware capital allocation changes the portfolio decision itself, not only the measured risk of a fixed book.

8 Future Work and Lessons Learnt

The most direct future work is to replace the synthetic exposure structure with a more realistic region-weighted or institution-specific portfolio. A second extension would be to strengthen the region calibration further so that DAPS and DiRe separate more clearly. A third would be to add formal confidence intervals to the Monte Carlo outputs.

Two lessons from the thesis are likely to be useful beyond this project. First, when a model changes several layers at once, a comparison ladder is essential; otherwise it becomes hard to tell whether a result changed because of the risk engine, the allocation rule, or the scenario system. Second, when a scenario is meant to describe geographic divergence, a sector-only portfolio can flatten the result in misleading ways. That lesson is directly supported by the improvement seen after the region-aware redesign.

9 Acknowledgments

I thank my supervisor for guidance throughout the thesis process and Prof. Nazim Madhavji for coordinating the undergraduate thesis course.

A Appendix

Supplementary material that would otherwise crowd the main body of the report is kept in the companion notebook and repository artifacts. In particular, the full walkthrough notebook contains the full code listing and implementation commentary, while the repository outputs directory contains the wider set of comparison charts and tables used to support the main report.

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